

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Semester Final Examination, Spring 2022
Course Code: CSE-3637 Course Title: Software Engineering
Time: 2.5 hours Marks: 50

[Answer any two questions from Group-A & any three from Group-B; separate answer scripts must be used for Group-A & Group-B]

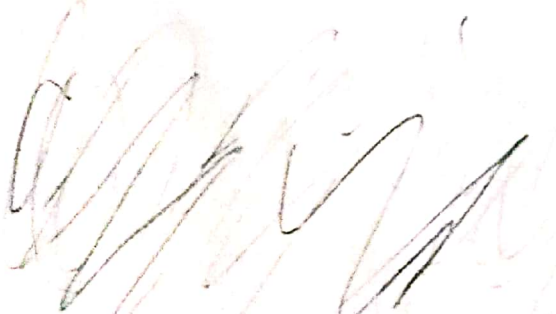
Group-A

- 1.a) Explain Top-Down & Bottom-Up approaches to software design. What are the different activities performed by user interface design? 5 CO4
- 1.b) Differentiate between coupling & cohesion with examples. What are the different types of software user interfaces? 5 CO3
- 2.a) As a registered user of the IIUC Student's Panel, who do you think are the stakeholders of this? Explain your answer in light of the categories of stakeholders. 5 CO4
- 2.b) Describe the standard for software documentation. 5 CO4
- Or
- 2.a) State and illustrate of design process function oriented design and object oriented design 5 CO4
- 2.b) Write short notes on: i) software design vs program design ii) software design vs software analysis iii) requirement engineering process. CO4

Group-B

- 4.a) How to design a Test Case? Point out the roles & responsibilities of a software tester. 5 CO4
- 4.b) Define stress testing with an example. After developing an application, Nikita asked some of her friends to test the app for her. Her friends tested the app, found some bugs, and reported them to Nikita. Which software testing technique represents the above scenario? Justify your answer. 5 CO4
- 5.a) How can software maintenance be defined? What are the types of software maintenance? Describe. 5 CO3
- 5.b) Depict the pathway of Software re-engineering considering program restructuring, forward engineering and reverse engineering. 5 CO3
- 3.a) Express Key Process Areas in capability maturity model. 5 CO3
- 3.b) Describe constructive cost estimation model. Define three types of projects in COCOMO model. 5 CO3

Or

- 3.a) Define software complexity. Suppose the fan out of a module is: 5, the cumulative number of variables passed to and from the module is: 9 and the complexity of the module is: 35. Use Card and Glass's Systems Complexity method to find out the structural complexity of the module. 5 CO3
- 3.b) Write short note on Delphi cost estimation, Heuristic technique, and project staffing plan 5 CO3
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International Islamic University Chittagong

Center for General Education (CGED)

Semester End Examination, Spring-2022

Course Code: URED-3604 Course Title: Life and Teachings of Prophet (SAAS)

(For Law faculty: URED-3201)

Full Marks: 50

Time: 2:30 Hours

Answer all questions strictly
(All questions are of equal value)

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	"The migration of the Prophet (SAAS) from Makkah to Madinah was a turning point in Islamic history"-assess the statement focusing on its lessons and significance.	10	2	Evaluate
2	a) Analyze the Battle of Badr mentioning its background, events and far-reaching lessons drawn from it. Or, b) Discuss the "Battle of <i>Ahzab</i> " describing the strategies taken by the Prophet Muhammad (SAAS) for achieving the grand success in this battle.	10	3 3	Analyze Understand
3	The <i>Hudaibiyyah</i> agreement is known to be as <i>Fathum Mubeen</i> "Clear Victory" in the Holy Qur'an, why? Discuss with proper evidence.	10	2	Understand
4	"The conquest of Makkah without any bloodshed is a remarkable event in the history of Islam"- Justify	10	3	Evaluate
5.	Evaluate the Farewell Address of Prophet Muhammad (SAAS) as an acknowledged document of human rights.	10	1,2	Evaluate

International Islamic University Chittagong
Morality Development Program (MDP)
Semester Final Examination, Spring 2022
Faculty of Science and Engineering, Semester- 6th

Course Title: Islamization of Discipline

Course Code: MDP-3606

Time: 2 Hours

Total Marks-50

Answer any *Five* questions; Figures in the right hand margin indicate full marks

- | | | |
|---|---|-----|
| 1 | a) What do you mean by Finger Print of human being. | 2 |
| | b) How pain receptors work in the human body? | 2 |
| | c) What are the Ethical issues in Computer Use? Write short notes on it from Islamic Perspective | 6 |
| 2 | a) Ibn Abbas radi Allahu anhu reported that the Prophet (peace be upon him) said: "Prioritize five tasks before doing five tasks"- what are they? | 5 |
| | b) According to Islamic approach, how to maintain good health | 5 |
| 3 | a) "Algebra" is named after which book? Who is the author of the book? | 2.5 |
| | b) Who is called the doctor of all doctors? Who is the father of Chemistry | 2.5 |
| | c) Name two Muslim scholars and their contributions in Mathematics and Geography. | 5 |
| 4 | a) Where the Lot Sea is located? What was the crime of Lot's tribe? | 4 |
| | b) What type of natural disaster seized the nation of Mosa (A), tribe of Nooh (A) and Qaroon? What was their offense? | 6 |
| 5 | What are the negative impacts of sex outside of marriage, homosexuality, pornography and short dress on society? | 10 |
| 6 | a) Mention the effect of timely prayers on human health | 5 |
| | b) Discuss the integrated impact of Fasting on society. | 5 |
| 7 | a) Explain Miracle from an <i>Islamic perspective</i> | 5 |
| | b) "Different messengers have different types of Miracle"- Explain three of them | 5 |

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Final Examination, Spring 2022
Course Code: CSE2525 Course Title: Data Communication

Total marks: 50

Time: 2 hours 30 minutes

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

- | | | M | C | U/L |
|------|--|---|---|-----|
| 1.a) | A corporation has a medium of satellite channel with a 1 MHz bandwidth. The corporation needs to create 4 separate independent channels each capable of sending at least 1 Mbps. You are to design an appropriate modulation technique. | 5 | 3 | A |
| 1.b) | Multiplexing and demultiplexing are paired techniques used together. Suppose a multiplexing system takes 4 input channels of 10 kbps using a time slot of 1-bits and produces the bitstream as found in "01010101". Design an appropriate multiplexing system with diagrams and find the frame rate, frame duration, input bit duration, output bitrate, output bit duration. | 5 | 3 | U |
| | OR, | | | |
| | Show Binary Amplitude Shift Keying and Binary Frequency Shift keying for the data 11110. Assume that the data travels in 1 second. | 5 | 3 | U |
| 2.a) | We need to use synchronous TDM and combine 20 digital sources, each of 100 Kbps. Each output slot carries 2 bits from each digital source, but one extra bit is added to each frame for synchronization. Answer the following questions:
i. What is the size of an output frame in bits?
ii. What is the output frame rate?
iii. What is the duration of an output frame?
iv. What is the output data rate?
v. What is the efficiency of the system (ratio of useful bits to the total bits). | 5 | 3 | U |
| 2.b) | Describe the working procedure of circuit switch network with necessary figure. | 5 | 3 | U |
| | OR, | | | |
| | Explain Amplitude Modulation with figure. Why are guard bands required between AM radio stations? | 5 | 3 | U |
| 3.a) | Design a parity check code where data word are 4 bits and codeword are 4 bits. How do you detect errors with this encoding using even parity? What are the limitations of this C(4, 3) scheme? | 5 | 3 | A |
| 3.b) | Explain with proper diagrams what would happen if appropriate window size were not taken for both go-back-N and ARQ. | 5 | 3 | A |
| | Or | | | |
| | Explain with proper diagrams what would happen if a frame were lost, an acknowledgement is lost, and consecutive acknowledgements are lost for selective repeat ARQ flow control | 5 | 3 | A |
| 4.a) | What is the difference between a basic service set (BSS) and an extended service set (ESS) defined by IEEE 802.11? Explain Hidden and Exposed Station Problems. | 5 | 2 | U |
| 4.b) | What is the difference between a unicast, multicast and broadcast address? If an Ethernet destination address is 07-01-02-03-04-05, what is the type of the address (unicast, multicast, or broadcast)? | 5 | 3 | U |
| | Or | | | |
| | Suppose the data bits are "10010101", then find the codeword for correcting single bit error using Hamming code. | 5 | 3 | A |
| | Write short notes of the followings (any three, 3 x 2) | | | |
| | 1. DVB-S, 2. P2P, 3. Wired LAN, 4. Wireless LAN | 6 | 3 | U |
| | 5. The satellite that is located at approximately 35,786 km above the Earth? | 4 | 2 | A |
| | 6. What is the reason of this error? | | | |
| | 7. The error rate is 10 ⁻⁶ per bit. | | | |

International Islamic University Chittagong
 Department of Computer Science and Engineering
 B Sc. in CSE

Final Examination, Spring-2022

Course Code: CSE-3635

Course Title: Artificial Intelligence

Total marks: 50

Time: 2 hours and 30 minutes

[Answer the following questions. The figures in the right-hand margin indicate full marks.]

GROUP A

1. a) Define forward chaining and backward chaining. CO1 2
 Or

Differentiate between Modus Ponens and Generalized Modus Ponens. CO1 2

- b) Following are the rules and initial facts in the rule base and database of facts of a rule based expert system. CO2 2

R1: IF A and C THEN F R2: IF A and E then G R3: IF B then E R4: IF G then D	Initial facts in the database A,B(both are true)
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Prove if A and B true then D is true by using Forward chaining inference procedure.

- c) Consider the following game tree.

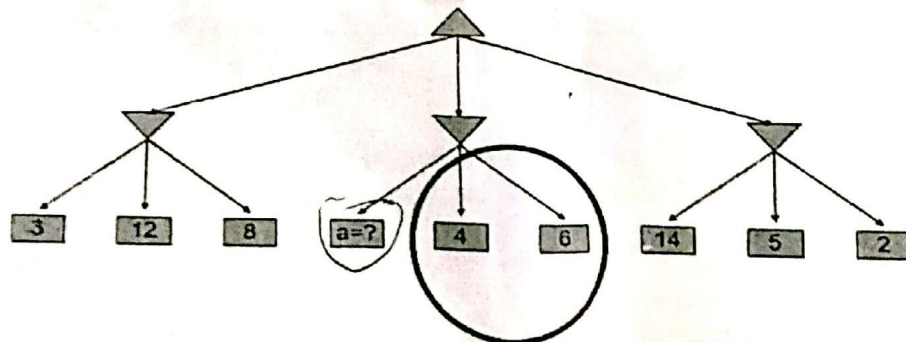
▲ Max

▼ Min

■ Terminal node

CO4 6

Notice that one terminal node marked 'a' has no value assigned (? marked). Suppose that the black circled node is pruned when you apply minimax search algorithm with alpha-beta pruning. Assign appropriate values to 'a' that will result in such a pruning.



2. a) Briefly discuss the steps through which NLP is conducted. CO1 2
 b) Write down the top down and bottom-up parse tree for the sentence "Book the flight through Houston" using the following CFG. CO3 5

Grammar

- S → NP VP
- S → Aux NP VP
- S → VP 2
- NP → Pronoun
- NP → Proper-Noun
- NP → Det Nominal
- Nominal → Noun
- Nominal → Nominal Noun
- Nominal → Nominal PP
- VP → Verb
- VP → Verb NP
- VP → VP PP
- PP → Prep NP

Lexicon

- Det → the | a | that | this
- Noun → book | flight | meal | money
- Verb → book | include | prefer
- Pronoun → I | he | she | me
- Proper-Noun → Houston | NWA
- Aux → does
- Prep → from | to | on | near | through

Or Some Propositional Rules and a KB for the Wumpus world are given.

CO3 5

Propositional rules:

$$(R1) \neg S_{11} \Rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$$

$$(R2) \neg S_{21} \Rightarrow \neg W_{11} \wedge \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$$

$$(R3) \neg S_{12} \Rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{22} \wedge \neg W_{13}$$

$$(R4) S_{12} \Rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$$

KB:

$\neg W_{11}$	$\neg S_{11}$	$\neg P_{11}$	$\neg B_{11}$	$\neg G_{11}$	V_{11}	OK_{11}
$\neg W_{12}$	—	$\neg P_{12}$	—	—	$\neg V_{12}$	OK_{12}
$\neg W_{21}$	$\neg S_{21}$	$\neg P_{21}$	B_{21}	$\neg G_{21}$	V_{21}	OK_{21}

Prove that Wumpus is in the room (1, 3)

c) Explain with example why universal quantification elimination is easy while existential quantification elimination is not that much easy.

Or

Discuss a semantic model to address the following sentences, where the word "bank" carries different meaning.

- i. I went to the bank to observe sunset.
- ii. I went to the bank to withdraw money.

Group B

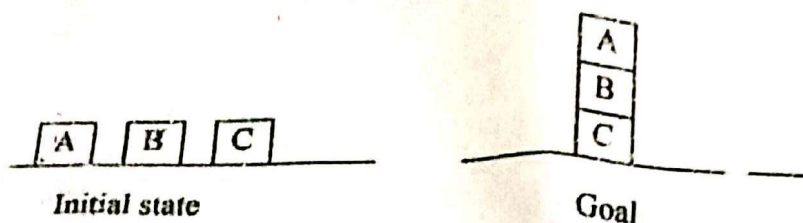
3. a) Write down the preconditions of the following operators:-

CO3 4

- i. ~~UNSTACK~~ (X, Y)
- ii. STACK (X, Y)
- iii. PICKUP (X)
- iv. PUTDOWN (X)

b) Find out the desired sequence of actions for achieving the goal in the following example.

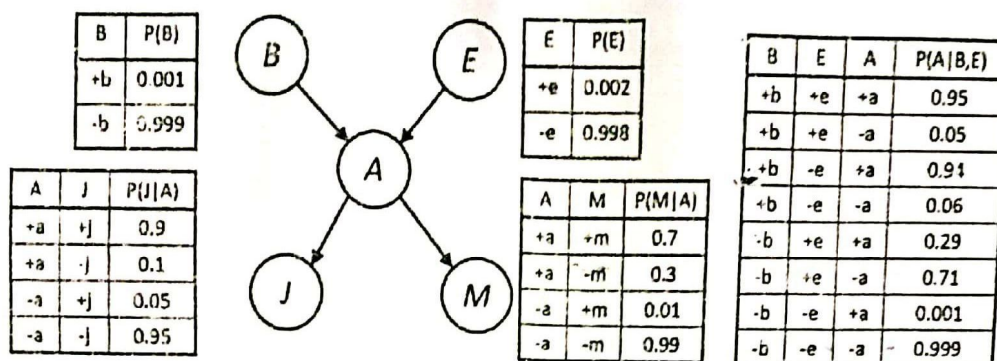
CO3 5



4. ~~a)~~ What is Baye's theorem?
Or What is inductive learning?

CO1 2

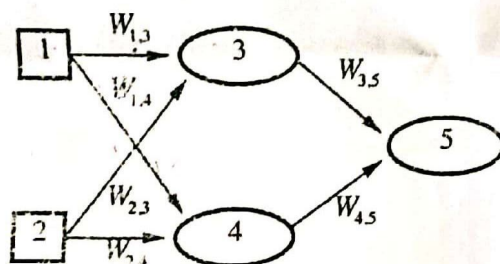
- b) The following figure is a Bayesian network showing both the topology and conditional probability tables. CO4 8



Calculate $P(+b|+j,+m)$ using inference by enumeration.

O Review the following neural network

R



$1 = 15, 2 = 8, W_{1,3} = 0.5, W_{1,4} = 0.3, W_{2,3} = 0.2, W_{2,4} = 0.4, W_{3,5} = 0.5, W_{4,5} = 0.4$

I. Calculate hidden layer value

II. Compute the value of Z without the transfer function.

5. a) Differentiate between supervised and unsupervised learning with examples. CO2 2

- b) The doctor knows that the disease meningitis causes the patients to have stiff neck, say, 40% of time. The doctor also knows the following facts, where, m: meningitis, S: stiff neck. CO3 6

$$P(+m) = 0.0001$$

$$P(+s|+m) = 0.8$$

$$P(+s|-m) = 0.01$$

Calculate $P(+m|+s)$.

- c) Let X is a set numbers near to zero. Define fuzzy and crisp sets for Z and show the triangular membership function curve in a graph. CO3 2

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Final Examination, Spring 2022
Course Code: CSE 3631 Course Title: Operating Systems

Total marks: 50

Time: 2 hours 30 minutes

The figures in the right hand margin indicate full marks.
 Course Outcomes and Bloom's Taxonomy Levels are mentioned in additional columns

Group A

CO DL

1.
 - a) What are the necessary conditions for a deadlock to occur? Name these conditions and provide an elaboration.
 - b) Consider the following snapshot of a system:

3 CO2 U

7 CO2 Ap

	Allocation				Request				Available			
	R ₁	R ₂	R ₃	R ₄	R ₁	R ₂	R ₃	R ₄	R ₁	R ₂	R ₃	R ₄
A	1	0	1	1	0	1	0	0	0	0	0	0
B	1	1	0	0	0	1	1	0				
C	1	0	1	1	0	0	0	0				
D	0	0	0	0	1	0	1	0				

Is the system in a deadlock state? If the system is in a deadlock state, list all processes that involve in a deadlock. Show your computation step-by-step.

2.
 - a) Compare the memory organization schemes of contiguous memory allocation and paging with respect to the following issues:

2 CO3 U

1. External fragmentation
2. Internal fragmentation

- b) Memory partitions of 100 MB, 170 MB, 205 MB, 300 MB, and 185 MB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 200 MB, 15 MB, 185 MB, 75 MB, 175 MB, and 30 MB (in order)? Indicate which—if any—requests cannot be satisfied. Comment on how efficiently each of the algorithms manages memory.

5 CO3 An

- c) Consider a logical address space of 2,048 pages with 4-KB page size, mapped onto a physical memory of 312 frames.
 1. How many bits are required in the logical address?
 2. How many bits are required in the physical address?

3 CO1 U

2.
 - a) Define Compaction and explain why it is used.
 - b) With the help of an example explain the paging concept.
 - c) Given memory holes (i.e., unused memory blocks) of 100K, 500K, 200K, 300K and 600K (in address order), how would each of the first-fit, best-fit and worst-fit algorithms allocate memory requests for 200K, 100K, 400K and 50K (in this order)?

3 CO1 U

3 CO3 An

4 CO3 Ap

3.
 - a) Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.

3 CO1 U

b) Consider the following page reference string:

7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1.

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms? Compare this two algorithms with the benchmark optimal page replace algorithm.

- LRU replacement
- FIFO replacement

7 CO3 Ap

4.

a) Illustrate how File-Allocation Table works.

2 CO1 U

b) Consider a file currently consisting of 100 blocks. Assume that the file-control block (and the index block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked, and indexed (single-level) allocation strategies, if, for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory.

5 CO3 An

1. The block is added at the beginning.
2. The block is added in the middle.
3. The block is added at the end.

c) Explain the paging hardware with TLB.

3 CO1 U

5.

a) Explain the terms 'WORMS' and 'VIRUSES' with reference to system threats.

3 CO5 U

OR

Analyze the principle of least privilege regarding system protection.

b) How do computer worms work?

3 CO5 An

OR

Explain the security violation categories with example.

c) Consider the following segment table:

4 CO3 Ap

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

- i. 0,430
- ii. 1,10
- iii. 2,500
- iv. 3,400

OR

Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, then calculate the effective memory access time (in milliseconds).

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering

Final Examination

Program: **B.Sc in CSE**

Course Code: **ECON-3501**

Total Marks: **50**

Semester: **Spring 2022**

Course Title: **Principles of Economics**

Time: **2 Hours 30 Minutes**

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

PART A

- | | | | | | |
|----|----|--|------|----|---|
| 1. | a) | Describe different types of market structure. | CLO1 | E | 5 |
| | b) | Given the following cost function $TC = 500 + 35Q - 7Q^2 + Q^3$
Calculate ATC, AVC and MC when the firm produces 40 units of output. | CLO2 | Ap | 5 |
| 2. | a) | How do we measure the GDP growth? Explain with an example. | CLO2 | An | 5 |
| | b) | What is Consumer Price Index? How it is used to measure inflation? | CLO3 | An | 5 |
| OR | | | | | |
| 2. | a) | What do you mean by nominal GDP and real GDP? Explain the benefit of measuring real GDP. | CLO2 | An | 5 |
| | b) | Suppose, GDP = \$5700, Net Factor payment from abroad = \$18
Capital consumption allowance = \$625, Indirect taxes = \$475
Social security contribution = \$530, Govt. and business transfers to person = \$770
Dividends = \$135, Personal Tax & Non-Tax payment = \$620
Required: Calculate GNP, NNP, National Income, Personal Income and Personal Disposable income | CLO3 | An | 5 |

PART B

- | | | | | | |
|----|----|--|------|----|----|
| 3. | a) | Discuss monetary policy and fiscal policy. How fiscal policy plays an important role in a developing country like Bangladesh? | CLO1 | E | 5 |
| | b) | Briefly explain unemployment with its classification. Evaluate the measures that can be taken to reduce unemployment problem in a country like Bangladesh. | CLO3 | U | 5 |
| 4. | a) | Discuss about the importance of economic planning of a country. | CLO3 | An | 5 |
| | b) | What are the economic role of the state? | CLO3 | An | 5 |
| 5. | | Write Short Notes On: Poverty in Bangladesh, Economic benefit of Karnofuli Tunnel, 8 th 5 year plan of Bangladesh, Unemployment | CLO1 | U | 10 |
| OR | | | | | |
| 5. | | Importance of Zakat, Value Added Approach of measuring GDP, Development Planning, Welfare vs GDP | CLO1 | U | 10 |